

The $RO(C_3)$ -Graded Cohomology of a Complete Flag Variety

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Abstract

Let C_3 act on the complete flag variety of its complex regular representation, and let R be the $RO(C_3)$ -graded cohomology of a point with constant Mackey functor coefficients $\mathbb{F}_3$. We compute the genuine cohomology ring as $R[x_1, x_2, x_3]/(e_1, e_2 + a^2, e_3)$, where a is the Euler class of the two-dimensional real irreducible representation λ and each x_i has degree λ . A splitting into six suspended $H\mathbb{F}_3$ -modules gives a comparison isomorphism with Borel cohomology in pure multiples of λ . This determines normalized Schubert lifts and their multiplication, including Euler corrections to the ordinary divisor formulas. We determine the kernel and image of restriction to the six fixed flags. The kernel is a nonzero square-zero ideal at the fixed-orbit level. If Γ is the module of tuples satisfying the root-curve edge congruences, then $\Gamma/\text{im } i^* \cong ((a)/(a^2))^{\oplus 2} \oplus (a)/(a^3)$ as graded R -modules. We also specify restriction and transfer in every representation degree.

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1 Introduction

The cohomology of a complete flag variety admits presentations in terms of Chern classes and Schubert classes. For a finite group acting on a flag variety, genuine equivariant cohomology is graded by virtual representations. Nonzero cohomology classes can vanish under the maps to Borel cohomology and fixed-point cohomology, so a Borel presentation alone does not determine the genuine ring.

We consider the complete flag variety of the complex regular representation of C_3 . Set

$$G = C_3 = \langle g \rangle, \quad V = \mathbb{C}[C_3] = \mathbf{1} \oplus \chi \oplus \chi^2, \quad X = \text{Fl}(V),$$

where $\chi(g) = \exp(2\pi i/3)$. Let $\lambda = \chi_{\mathbb{R}}$, $H = H\mathbb{F}_3$, and

$$R^\alpha = \widetilde{H}_G^\alpha(S^0; \mathbb{F}_3) = \pi_{-\alpha}^G H, \quad R = R^\star.$$

Throughout, $H_G^\star(Y)$ means unreduced cohomology of an unbased space, or equivalently reduced cohomology of Y_+ . The superscript $\star$ ranges over $RO(G) = \mathbb{Z}\{1, \lambda\}$. Write $\underline{H}_G^\star(Y)$ for the graded cohomology Mackey functor and $\underline{R}^\star = \underline{H}_G^\star(\text{pt})$. Without an underline, cohomology denotes its value at G/G . Write $a = e(\lambda) \in R^\lambda$.

Theorem 1.1. *There are classes $x_i \in H_G^\lambda(X)$ for which*

$$H_G^\star(X) \cong \frac{R[x_1, x_2, x_3]}{(e_1(x), e_2(x) + a^2, e_3(x))}, \quad |x_i| = \lambda. \quad (1.1)$$

Here e_j denotes the j th elementary symmetric polynomial. As an R -module the ring is free on $1, A, B, C, D, F$, of respective degrees $0, \lambda, \lambda, 2\lambda, 2\lambda, 3\lambda$, where

$$\begin{aligned} A &= x_1, & B &= x_1 + x_2 + a, & C &= x_1 x_2, \\ D &= x_1(x_1 + a), & F &= x_1 x_2(x_1 + a). \end{aligned} \quad (1.2)$$

These are lifts of the ordinary Schubert classes indexed by $s_1, s_2, s_1 s_2, s_2 s_1, w_0$, respectively. Equivalently,

$$H_G^\star(X) \cong \frac{R[A, B]}{(A^3 - a^2 A, A^2 - AB + B^2 + a(A + B))}. \quad (1.3)$$

The theorem uses the full representation-graded coefficient ring, including its negative cone. The generators are defined by a comparison isomorphism in degree λ ; no genuine projective bundle theorem or general complex orientation is assumed. The first Schubert products are

$$A^2 = D - aA, \quad B^2 = C - aB, \quad AB = C + D. \quad (1.4)$$

The terms involving a vanish after forgetting the action. Table 3 gives the remaining products.

The six fixed flags are indexed by S_3 . The restriction map

$$i^* : H_G^\star(X) \longrightarrow R^6$$

has a nonzero kernel. With $J_r = \text{Ann}_R(a^r)$, we prove

$$\ker i^* = J_1 A \oplus J_1 B \oplus J_2 C \oplus J_2 D \oplus J_3 F. \quad (1.5)$$

This ideal has square zero at G/G . Restriction becomes an isomorphism after inverting a . Before localization, five congruences describe its image: two modulo a , two modulo a^2 , and one modulo a^3 . Let $\Gamma \subset R^6$ be the graded module defined by the root-curve edge conditions. Then

$$\Gamma / \text{im } i^* \cong \left(\frac{aR}{a^2R} \right)^{\oplus 2} \oplus \frac{aR}{a^3R}. \quad (1.6)$$

The tuple $(0, 0, 0, 0, 0, a)$ represents a nonzero class in the last summand, in a degree where restriction itself is injective.

A Morse filtration with the Schubert representation shifts gives an H -module splitting and hence a comparison with Borel cohomology in degrees $n\lambda$. A basis argument then proves the ring presentation over the full coefficient ring R .

Relation to previous work

Lewis computed representation-graded cohomology of projective spaces with linear C_p -actions using Burnside coefficients [10]. Under the cell hypotheses of her Theorem 3.3, Shulman gives a corrected proof of joint detection by underlying cohomology and fixed-point restriction in even underlying degrees [12, Theorems 3.10–3.11]. That result uses Burnside coefficients over a ground ring without p -torsion and does not apply to our constant $\mathbb{F}_3$ coefficients. Basu and Ghosh compute the latter coefficient ring for C_p and prove freeness for sparse representation-cell complexes [2, §3.2 and Theorem 4.12]. Our cells have real dimensions 0, 2, 2, 4, 4, 6, so their sparseness condition holds. We use their coefficient calculation to prove the H -module splitting and its representation shifts directly.

Basu, Dey, and Karmakar prove $H\mathbb{Z}$ -module splittings for projectivizations of representations, including $\mathbb{P}(V)$ [3, Theorem 4.7]. Extension of scalars to H gives shifts $0, \lambda, 2\lambda$ in this case. Their projective-space ring calculations include Euler corrections [3, Example 7.5 and Theorem 7.30]. The cubic relation $x_1^3 = a^2x_1$ has a precedent in these projective-space calculations; the flag calculation also determines the mixed quadratic relation and its Schubert products. Although $X \cong \mathbb{P}(\mathcal{Q})$ for the tautological quotient $\mathcal{Q} = (\mathbb{P}(V) \times V)/L$, the cited results concern projectivizations of representations over a point, not arbitrary equivariant bundles. Here L is the tautological line on $\mathbb{P}(V)$, and E_1, E_2, E_3 are the summands $\mathbf{1}, \chi, \chi^2$ of V , respectively. The real fibers of $\mathcal{Q}$ at E_1, E_2, E_3 have types $2\lambda, 2 + \lambda, 2 + \lambda$, respectively. We therefore construct the required degree- λ classes by comparison and use an ordinary projective bundle calculation only after passing to Borel cohomology. Basu and Dey give coefficient-ring computations for cyclic groups with constant integral and Burnside coefficients [1].

Genuine flag-space calculations also occur in Kronholm’s dissertation [9, Proposition X.2]: the spaces there are finite-length real flags in a complete C_2 -universe, with constant $\mathbb{F}_2$ coefficients. For the classical ordinary and torus-equivariant Schubert constructions used below, we follow the conventions in [4, §1.2] and [8, §2]. We compute the cohomology of this finite flag variety over the full coefficient ring, including its Schubert basis and the kernel, image, and edge-congruence defect of fixed-point restriction. The splitting and comparison arguments use the known coefficient calculation. The GKM theorem of Harada, Henriques, and Holm [6, Theorem 3.1] requires non-zero-divisor Euler classes and a relative-primality condition. Both Euler-class conditions fail here, as explained in Section 8; the defect computation does not contradict that theorem.

Sections 2–4 establish the coefficient conventions and the splitting. Section 5 proves the comparison statement and computes the Borel ring. Sections 6 and 7 prove Theorem 1.1 and compute Schubert products. Section 8 treats fixed-point restriction, and Section 9 gives the Mackey structure and ordinary specializations.

2 Conventions and the coefficient ring

We use cohomological representation degrees. The positive coefficient classes have degrees

$$|a| = \lambda, \quad |u| = \lambda - 2, \quad |\kappa| = \lambda - 1. \quad (2.1)$$

Choose u using the complex orientation of χ on restriction to the trivial subgroup. The same letter will denote its image in Borel cohomology. The element u is not invertible in R .

In our grading, the coefficient calculation in [2, Definition 3.3 and Proposition 3.4, pp. 39–41] takes the form

$$R = P \oplus N, \quad P = \mathbb{F}_3[a, u] \otimes \Lambda(\kappa), \quad N = \bigoplus_{j,k \geq 1} \mathbb{F}_3\{\nu_{j,k}, \mu_{j,k}\}, \quad (2.2)$$

where

$$|\nu_{j,k}| = 1 + 2j - (j + k)\lambda, \quad |\mu_{j,k}| = 2j + (1 - j - k)\lambda. \quad (2.3)$$

We normalize the symbols so that

$$\begin{aligned} a\nu_{j,k} &= \nu_{j,k-1}, & u\nu_{j,k} &= \nu_{j-1,k}, & a\mu_{j,k} &= \mu_{j,k-1}, & u\mu_{j,k} &= \mu_{j-1,k}, \\ \kappa\nu_{j,k} &= \mu_{j,k}, & \kappa\mu_{j,k} &= 0. \end{aligned} \quad (2.4)$$

A symbol with a zero index means zero. Also $N^2 = 0$. The last relation follows from the module description: every element of N is infinitely divisible by a and killed by a power of a . If $a^r n = 0$ and $n' = a^r m$ with $m \in N$, then $nn' = 0$. The symbols in (2.3) correspond to the suspended fractional symbols in the cited homotopy calculation after reversing degrees. They are not fractions in R .

For $\alpha = m + n\lambda$, the parity is that of its underlying dimension $m + 2n$. Thus a and u are even, whereas κ and $\nu_{j,k}$ are odd. All polynomial generators of the flag ring commute with the coefficients.

Lemma 2.1. *In the above convention,*

$$R^{1-r\lambda} = 0 \quad (r \geq 0), \quad R^{r\lambda} = \begin{cases} \mathbb{F}_3 a^r, & r \geq 0, \\ 0, & r < 0. \end{cases} \quad (2.5)$$

For $r \geq 1$,

$$J_r = \text{Ann}_R(a^r) = \text{span}_{\mathbb{F}_3}\{\nu_{j,k}, \mu_{j,k} : j \geq 1, 1 \leq k \leq r\}. \quad (2.6)$$

Proof. A monomial $a^i u^j \kappa^\epsilon$ in P has degree $(-2j - \epsilon) + (i + j + \epsilon)\lambda$, with $i, j \geq 0$ and $\epsilon \in \{0, 1\}$. Its coefficient of the trivial representation cannot be 1, and can be 0 only when $j = \epsilon = 0$. For a negative-cone generator, this coefficient is at least 2. This proves (2.5). Multiplication by a is injective on P . On N , its powers shift the second index as in (2.4), giving (2.6). $\square$

Choose $s \in H^1(BG; \mathbb{F}_3)$ so that $t = \beta s$ is the first Chern class of the line bundle associated to χ . Put

$$R_B^* = \widetilde{H}_G^*(EG_+) = H^*(BG; \mathbb{F}_3)[u^{\pm 1}] = \mathbb{F}_3[t] \otimes \Lambda(s)[u^{\pm 1}], \quad |s| = 1, \quad |t| = 2.$$

The coefficient comparison, with this orientation, is

$$q_R : R \longrightarrow R_B, \quad a \longmapsto ut, \quad u \longmapsto u, \quad \kappa \longmapsto us, \quad N \longmapsto 0. \quad (2.7)$$

This is the calculation in [2, §3.2, pp. 40–41]. In particular,

$$R_B = R[u^{-1}], \quad \ker q_R = N, \quad R[a^{-1}] = \mathbb{F}_3[a^{\pm 1}, u] \otimes \Lambda(\kappa). \quad (2.8)$$

For example $0 \neq \mu_{1,1} \in R^{2-\lambda}$ is killed by both a and u . Section 9 gives the coefficient restriction and transfer maps.

3 Equivariant Schubert cells

Fix a basis e_1, e_2, e_3 of V on which $g = \text{diag}(1, \zeta, \zeta^2)$, where $\zeta = \exp(2\pi i/3)$, and write $E_i = \mathbb{C}e_i$. We compose permutations as functions: $(vw)(i) = v(w(i))$. In one-line notation,

$$s_1 = 213, \quad s_2 = 132, \quad s_1 s_2 = 231, \quad s_2 s_1 = 312, \quad w_0 = 321.$$

Write $\ell(w)$ for the number of inversions of w . All tables use the order 123, 213, 132, 231, 312, 321.

3.1 Fixed flags and tangent representations

The three eigenvalues of g are distinct. Its spectral projections are polynomials in g , so every invariant subspace of V is a sum of eigenlines. It follows that the fixed flags are exactly

$$p_w = (E_{w_1} \subset E_{w_1} \oplus E_{w_2} \subset V), \quad w \in S_3. \quad (3.1)$$

At such a flag the complex tangent representation is

$$T_{p_w} X \cong \bigoplus_{i < j} \text{Hom}(E_{w_i}, E_{w_j}) = \bigoplus_{i < j} \chi^{w_j - w_i}. \quad (3.2)$$

Every summand is nontrivial, and the underlying real representation is 3λ at each fixed point. Table 1 lists the complex weights.

The real representations underlying χ and χ^2 are isomorphic. Complex conjugation identifies them and reverses the complex orientation. Write e_B for the complex-oriented Borel Euler class; then $e_B(\chi) = t$ and $e_B(\chi^2) = -t$. Using the fixed u , we normalize these factors in degree λ as $q_R^{-1}(ut) = a$ and $q_R^{-1}(-ut) = -a$, where the inverse is taken in degree λ . These normalized factors differ from transporting canonical genuine Euler classes: conjugation sends $e((\chi^2)_{\mathbb{R}})$ to a , while sending the complex orientation class of $(\chi^2)_{\mathbb{R}}$ to $-u$. We use the normalized factors in all Euler products below.

3.2 Linear coordinates and the filtration

Let $\mathcal{B}$ be the upper triangular subgroup of $GL(V)$. The Schubert cell $\mathcal{B}p_w$ has unique echelon coordinates

$$v_i = e_{w_i} + \sum_{\substack{j > i \\ w_j < w_i}} z_{ji} e_{w_j}, \quad F_r = \langle v_1, \dots, v_r \rangle. \quad (3.3)$$

To obtain these coordinates one normalizes each pivot and subtracts earlier pivot columns; the remaining entries are precisely those indexed by inversions. Applying g and renormalizing each pivot gives

$$g \cdot z_{ji} = \zeta^{w_j - w_i} z_{ji}.$$

Thus the cell is equivariantly isomorphic, as an affine space, to the linear representation

$$W_w = \bigoplus_{\substack{i < j \\ w_i > w_j}} \chi^{w_j - w_i}. \quad (3.4)$$

For $w = 231$ the nonconstant columns are $e_2 + z_{31}e_1$ and $e_3 + z_{32}e_1$, with weights χ^2 and χ . For $w = 321$ they are $e_3 + z_{21}e_2 + z_{31}e_1$ and $e_2 + z_{32}e_1$, with weights χ^2, χ, χ^2 .

Table 1: Complex tangent weights and Schubert representations. The tangent weights are ordered by $(1, 2), (1, 3), (2, 3)$.

w	Tangent weights	W_w	$(W_w)_{\mathbb{R}}$
123	(χ, χ^2, χ)	0	0
213	(χ^2, χ, χ^2)	χ^2	λ
132	(χ^2, χ, χ^2)	χ^2	λ
231	(χ, χ^2, χ)	$\chi \oplus \chi^2$	2λ
312	(χ, χ^2, χ)	$\chi \oplus \chi^2$	2λ
321	(χ^2, χ, χ^2)	$\chi \oplus 2\chi^2$	3λ

The Bruhat closure relation gives a closed filtration $X_k = \bigcup_{\ell(w) \leq k} \mathcal{B}p_w$ for $0 \leq k \leq 3$; see [4, Proposition 1.2.1]. This is the restriction of the standard torus-equivariant Schubert representation-cell decomposition [6, §5], so the inclusions are equivariant cofibrations. Its successive quotients are the one-point compactifications of the cells of a fixed length:

$$X_k/X_{k-1} \cong \bigvee_{\ell(w)=k} S^{W_w} \cong \bigvee_{\ell(w)=k} S^{k\lambda}. \quad (3.5)$$

For $k = 0$ this means $(X_0)_+ \cong S^0$.

For the splitting argument, we use a Morse filtration with the same representation shifts. Represent a flag by its ordered orthogonal lines, choose unit vectors v_i on them, and let U be the unitary matrix with columns v_1, v_2, v_3 . Put

$$h(F) = 10\langle D_0 v_1, v_1 \rangle + 3\langle D_0 v_2, v_2 \rangle, \quad D_0 = \text{diag}(0, 1, 4). \quad (3.6)$$

Its critical-point equation is $[D_0, U \text{diag}(10, 3, 0)U^*] = 0$, so the critical points are the six p_w . Put $(c_1, c_2, c_3) = (10, 3, 0)$ and $(d_1, d_2, d_3) = (0, 1, 4)$. On the real root two-plane indexed by $i < j$, the Hessian has the sign of $(c_i - c_j)(d_{w_j} - d_{w_i})$. These factors are nonzero, so the critical points are nondegenerate and the negative representation is W_w . In our fixed-point order the critical values are 3, 10, 12, 22, 40, 43, with lengths 0, 1, 1, 2, 2, 3.

The equivariant Morse lemma and attachment theorem [13, Lemma 4.1 and Theorem 4.6] apply here. Compactness of X makes an invariant metric complete and ensures Wasserman's condition (C); the critical points are nondegenerate and have distinct critical values. The construction follows the ordinary Morse attachment argument [11, Theorem 3.2], using invariant gradient flow. Writing W_w^+ for the positive Hessian representation, the local handle pair $(D(W_w) \times D(W_w^+), S(W_w) \times D(W_w^+))$ retracts equivariantly to $(D(W_w), S(W_w))$ by contracting the positive directions. Thus X has the equivariant homotopy type of a finite representation-cell complex, obtained in the displayed order by attaching one $D(W_w)$ at a time. Its successive homotopy cofibers are $S^{\ell(w)\lambda}$. We use this filtration below.

4 Additive Bredon cohomology

Proposition 4.1. *There is an equivalence of genuine G -equivariant H -modules*

$$H \wedge X_+ \simeq H \vee (\Sigma^\lambda H)^{\vee 2} \vee (\Sigma^{2\lambda} H)^{\vee 2} \vee \Sigma^{3\lambda} H. \quad (4.1)$$

Consequently, as $RO(G)$ -graded Mackey functors,

$$\underline{H}_G^\alpha(X) \cong \bigoplus_{w \in S_3} \underline{R}^{\alpha - \ell(w)\lambda}. \quad (4.2)$$

Proof. Use the preceding representation-cell model, whose first stage is a fixed point. Suppose the preceding stage, after smashing with H , is a wedge of modules $\Sigma^{j\lambda}H$ with $j \leq k$. The cofiber of the next attachment is $\Sigma^{k\lambda}H$. Desuspending its connecting map gives components in

$$[\Sigma^{k\lambda-1}H, \Sigma^{j\lambda}H]_{H\text{-mod}}^G = R^{1-(k-j)\lambda} = 0. \quad (4.3)$$

The free rank-one H -module adjunction gives the equality, with $R^\alpha = \pi_{-\alpha}^G H$, and Lemma 2.1 gives the vanishing, including when $j = k$. The connecting map is therefore null in the genuine H -module category, so its cofiber triangle splits. Induction gives (4.1).

Apply the internal H -module dual $F_H(-, H)$. The free-module adjunction identifies $F_H(H \wedge X_+, H)$ with $F(X_+, H)$, and the dual of the finite wedge is the wedge of the $\Sigma^{-\ell(w)\lambda}H$. Taking all equivariant homotopy Mackey functors proves (4.2). $\square$

Equation (4.3) splits the H -module triangles and fixes the six representation shifts. Abstract R -module freeness alone would not identify those shifts. No splitting of X_+ into sphere spectra, and no multiplicative splitting, is asserted.

Equation (4.2) gives

$$\dim_{\mathbb{F}_3} H_G^{n\lambda}(X) = 1, 3, 5, 6, 6, \dots \quad \text{for } n = 0, 1, 2, 3, 4, \dots$$

The powers of a account for the difference from the ordinary Betti numbers 1, 2, 2, 1.

5 Comparison with Borel cohomology

Write $Y_B = EG \times_G Y$ and $H_B^*(Y) = H^*(Y_B; \mathbb{F}_3)$. We distinguish this integer-graded ring from

$$H_G^*(EG \times Y) = H_B^*(Y)[u^{\pm 1}].$$

The identification follows from the oriented representation-sphere suspensions on a free G -space [2, Proposition 3.2]. In particular its component of degree $n\lambda$ is $u^n H_B^{2n}(Y)$.

The next proposition follows from finite H -module duality and (2.7).

Proposition 5.1 (Comparison in pure representation degrees). *Let Y be a finite G -CW complex such that*

$$H \wedge Y_+ \simeq \bigvee_{j=1}^m \Sigma^{k_j\lambda} H, \quad k_j \geq 0. \quad (5.1)$$

For every integer n , the natural comparison is an isomorphism

$$q_Y : H_G^{n\lambda}(Y) \xrightarrow{\cong} u^n H_B^{2n}(Y). \quad (5.2)$$

These maps preserve products. Under hypothesis (5.1), the comparison on the entire $RO(G)$ -graded ring is localization at u , and

$$\ker q_Y = NH_G^*(Y). \quad (5.3)$$

Proof. Put $C_Y = F(Y_+, H)$. Dualizing (5.1) gives $C_Y \simeq \bigvee_j \Sigma^{-k_j\lambda} H$. Write $\eta_C : C \rightarrow F(EG_+, C)$ for precomposition with $EG_+ \rightarrow S^0$. Under $F(EG_+, C_Y) \simeq F((EG \times Y)_+, H)$, the map η_{C_Y} is the cochain pullback along $EG \times Y \rightarrow Y$. For any chosen splitting $\phi : C_Y \xrightarrow{\sim} C' = \bigvee_j \Sigma^{-k_j\lambda} H$, naturality gives

$$F(EG_+, \phi) \circ \eta_{C_Y} = \eta_{C'} \circ \phi.$$

Finite wedges are finite products, so $\eta_{C'}$ is the direct sum of the summand comparisons. Function spectra also commute with suspension by an invertible representation sphere. In cohomological degree $n\lambda$ the j th component is therefore

$$R^{(n-k_j)\lambda} \longrightarrow R_B^{(n-k_j)\lambda}.$$

For $r = n - k_j \geq 0$ both groups are one-dimensional, and the map sends a^r to $u^r t^r$. For $r < 0$ both groups vanish. This proves (5.2). The original comparison is a map of cohomology rings, so these isomorphisms preserve products.

In arbitrary degrees the same direct-sum calculation identifies the map with coefficient localization on each summand. Since the sum is finite, it induces an isomorphism after inverting u , with kernel the sum of the shifted copies of N . This sum is precisely $NH_G^*(Y)$. $\square$

In particular, (5.2) applies to X by Proposition 4.1. We use it to lift relations only in pure multiples of λ . Equation (5.3) shows that regrading Borel cohomology does not recover the full genuine ring.

5.1 The ordinary flag bundle

Let $\mathcal{E}_i = EG \times_G E_i$ over BG . Then X_B is the complete flag bundle of

$$\mathcal{V} = \mathcal{E}_1 \oplus \mathcal{E}_2 \oplus \mathcal{E}_3 = \mathbf{1} \oplus \mathcal{L}_\chi \oplus \mathcal{L}_{\chi^2}.$$

Let L_i denote the equivariant tautological line quotients on X and also their associated bundles on X_B . On the Borel flag bundle write $\mathcal{F}_i$ for the tautological subbundles, so that $L_i = \mathcal{F}_i / \mathcal{F}_{i-1}$, and put $z_i = c_1(L_i^\vee)$. The original bundle has roots $0, t, 2t = -t$, so

$$c(\mathcal{V}) = (1 + 0)(1 + t)(1 - t) = 1 - t^2.$$

Its dual has roots $0, -t, t$ and the same total Chern class. Hence

$$e_1(z) = 0, \quad e_2(z) = -t^2, \quad e_3(z) = 0. \quad (5.4)$$

Proposition 5.2. *There is an isomorphism of $H^*(BG; \mathbb{F}_3)$ -algebras*

$$H_B^*(X) = \frac{\mathbb{F}_3[t, z_1, z_2, z_3] \otimes \Lambda(s)}{(e_1(z), e_2(z) + t^2, e_3(z))}, \quad |z_i| = |t| = 2, \quad |s| = 1. \quad (5.5)$$

A module basis over $H^(BG; \mathbb{F}_3)$ is*

$$1, z_1, z_2, z_1 z_2, z_1^2, z_1^2 z_2. \quad (5.6)$$

Proof. First choose L_1 by passing to $\mathbb{P}(\mathcal{V})$, then choose L_2 by projectivizing the rank-two quotient $\mathcal{V}/L_1$. The classes $1, z_1, z_1^2$ and $1, z_2$ restrict to bases on the first and second projective fibers, respectively. These tautological classes are defined on the total spaces, so two applications of ordinary Leray–Hirsch [7, Theorem 4D.1], with $\mathbb{F}_3$ coefficients, give (5.6).

The Chern identities (5.4) give a map from the displayed quotient to cohomology. Eliminating z_3 identifies that quotient with the successive monic extensions

$$\left(H^*(BG; \mathbb{F}_3)[z_1]/(z_1^3 - t^2 z_1) \right) [z_2]/(z_2^2 + z_1 z_2 + z_1^2 - t^2).$$

It has exactly the basis (5.6), which maps to the Leray–Hirsch basis. The map is an isomorphism. $\square$

Since every basis element in (5.6) is even, the even part of (5.5) consists of the terms without s . We use this when defining the genuine generators.

6 The genuine cohomology ring

6.1 Generators and a basis argument

By Proposition 5.1 in degree λ , there is a unique class $x_i \in H_G^\lambda(X)$ satisfying

$$q_X(x_i) = uz_i = u c_1^B(L_i^\vee). \quad (6.1)$$

The degree is $|u| + 2 = \lambda$. The real fiber types of L_i at fixed flags vary between 2 and λ ; the definition above uses the comparison isomorphism and does not assume a Thom class for L_i in a constant representation degree. Naturality of comparison gives

$$\text{res}(x_i) = vy_i, \quad y_i = c_1(L_i^\vee) \in H^2(X; \mathbb{F}_3), \quad v = \text{res}(u). \quad (6.2)$$

Here v records degree $\lambda - 2$ at the trivial subgroup; after forgetting the representation grading it is the unit in degree zero.

The following lemma identifies a basis from classes in these degrees.

Lemma 6.1. *Suppose $b_1, \dots, b_m$ is a homogeneous R -basis of $H_G^*(Y)$ supplied by a splitting as in (5.1), with $|b_j| = k_j\lambda$. Let $c_j \in H_G^{k_j\lambda}(Y)$ have restrictions which, after removing the factors v^{k_j} , form an ordinary cohomology basis of Y . Then the c_j form an R -basis of $H_G^*(Y)$ and a free $\underline{R}$ -basis of the graded Mackey functor $\underline{H}_G^*(Y)$.*

Proof. Order both families by nondecreasing k_j . In $c_j = \sum_i d_{ij}b_i$, the coefficient d_{ij} lies in $R^{(k_j - k_i)\lambda}$. By Lemma 2.1, it is zero if $k_i > k_j$ and is a scalar multiple of $a^{k_j - k_i}$ otherwise. The change-of-basis matrix is therefore block triangular; its diagonal blocks have entries in $\mathbb{F}_3$.

The underlying $H\mathbb{F}_3$ -module splitting gives an ordinary basis from the b_i . Restriction kills the positive powers of a , so the diagonal blocks are precisely the ordinary degree-wise change-of-basis matrices. They are invertible over $\mathbb{F}_3$ by hypothesis. A finite block triangular matrix with invertible diagonal is invertible over R : after factoring out its diagonal part, invert the remaining $1 + T$ by the finite series $1 - T + T^2 - \dots$, since T is strictly block triangular. These operations are coefficient-module maps on both orbits. Equivalently they give an invertible matrix of maps between the free suspended H -modules dual to the chosen basis. Thus they also give a basis at the Mackey functor level. $\square$

6.2 Proof of the presentations

Proof of Theorem 1.1. Under comparison, the three expressions $e_1(x), e_2(x) + a^2, e_3(x)$ map to $ue_1(z), u^2(e_2(z) + t^2), u^3e_3(z)$, respectively. They vanish by Proposition 5.2. They have degrees $\lambda, 2\lambda, 3\lambda$, where comparison is injective. The genuine relations therefore hold.

Let Q denote the quotient in (1.1). After setting $x_3 = -x_1 - x_2$, put

$$f = x_2^2 + x_1x_2 + x_1^2 - a^2, \quad h = x_1^3 - a^2x_1.$$

Then $e_2 + a^2 = -f$ and $e_3 = h - x_1f$. Thus

$$Q \cong (R[x_1]/(h)) [x_2]/(f). \quad (6.3)$$

Division by a monic polynomial works over an arbitrary coefficient ring, including one with zero divisors. Successive division in (6.3) gives the unique module normal forms

$$1, x_1, x_2, x_1x_2, x_1^2, x_1^2x_2. \quad (6.4)$$

Their images in cohomology restrict, with the appropriate powers of v , to the ordinary basis obtained from the two projective bundles over a point. Lemma 6.1, applied to Proposition 4.1, shows that these images are an R -basis of $H_G^*(X)$. The map $Q \rightarrow H_G^*(X)$ therefore maps a basis to a basis and is an isomorphism in every representation degree.

Define A, B, C, D, F by (1.2). The remaining monomials are

$$x_1 = A, \quad x_2 = B - A - a, \quad x_1 x_2 = C, \quad x_1^2 = D - aA, \quad x_1^2 x_2 = F - aC.$$

Thus $1, A, B, C, D, F$ is another R -basis. Their Schubert normalization is proved geometrically in the next section. Finally substitute

$$x_1 = A, \quad x_2 = B - A - a, \quad x_3 = a - B.$$

The polynomial h becomes $A^3 - a^2 A$, and in characteristic three f becomes $A^2 - AB + B^2 + a(A + B)$. This invertible change of variables proves (1.3). $\square$

The generator products were detected in pure multiples of λ . The presentation holds over all of R , so (1.1) also includes multiplication by negative-cone coefficients, which q_X kills.

Let $\pi : X \rightarrow \mathbb{P}(V)$ select the first line. The standard filtration of $\mathbb{P}(V)$ has representations $0, \lambda, 2\lambda$. Applying the splitting, comparison, and basis arguments above to this filtration gives

$$H_G^*(\mathbb{P}(V)) = R[\eta]/(\eta^3 - a^2 \eta), \quad q_{\mathbb{P}(V)}(\eta) = u c_1^B(L^\vee), \quad \pi^* \eta = x_1.$$

Here the Borel relation is the ordinary projective bundle relation for roots $0, t, -t$. The classes $1, \eta, \eta^2$ restrict to the ordinary projective-space basis, so the same monic and triangular arguments prove the displayed isomorphism over all of R . Equation (6.3) then makes π^* injective and exhibits $H_G^*(X)$ as a free module on $1, x_2$ over this subring. The additional relation is $x_2^2 + x_1 x_2 + x_1^2 = a^2$. This describes the particular bundle $\mathbb{P}(\mathcal{Q})$ without requiring a general genuine projective bundle theorem.

7 Equivariant Schubert calculus

7.1 Geometric normalization

Use opposite Schubert varieties relative to the upper triangular cells of Section 3. Their complex codimensions are $\ell(w)$. Denote their Borel classes by S_w . The underlying ordinary classes form the usual Schubert basis [4, §1.3]; the analogous torus-equivariant basis is described in [8, Theorem 2.2]. Here the Borel representatives come directly from projections of the tautological bundles.

For s_1 the condition is $F_1 \subset E_2 \oplus E_3$. On the Borel flag bundle this is the zero locus of $L_1 \rightarrow \mathcal{E}_1$, so its divisor class is z_1 . For s_2 the condition is $E_3 \subset F_2$. It is the zero locus of the determinant of $\mathcal{F}_2 \rightarrow \mathcal{E}_1 \oplus \mathcal{E}_2$, giving $z_1 + z_2 + t$. These are regular divisor conditions.

For 231 the condition is $F_2 = E_2 \oplus E_3$, the regular zero locus of $\mathcal{F}_2 \rightarrow \mathcal{E}_1$. Its class is $c_2(\mathcal{F}_2^\vee \otimes \mathcal{E}_1) = z_1 z_2$. For 312 the condition is $F_1 = E_3$, the regular zero locus of $L_1 \rightarrow \mathcal{E}_1 \oplus \mathcal{E}_2$. Its class is $z_1(z_1 + t)$. These loci are smooth copies of a projective line in each fiber. On the latter locus, projection to $\mathcal{E}_1$ vanishes on $\mathcal{F}_1 = \mathcal{E}_3$ and hence descends to $L_2 = \mathcal{F}_2/\mathcal{F}_1$. The condition $F_2 = E_2 \oplus E_3$ is the regular zero locus of $L_2 \rightarrow \mathcal{E}_1$, with class z_2 . The projection formula gives the class $z_1 z_2(z_1 + t)$ for the opposite Schubert point 321. This also fixes the complex orientations used for these classes.

Definition 7.1. The genuine Schubert lift $\sigma_w \in H_G^{\ell(w)\lambda}(X)$ is the unique class with $q_X(\sigma_w) = u^{\ell(w)} S_w$.

Proposition 5.1 gives existence and uniqueness. The geometric calculations above and (6.1) give Table 2, identifying the basis in the main theorem with the normalized Schubert basis.

7.2 Products

Proposition 7.2. *The multiplication of the nonunit Schubert classes is given in Table 3. Together with R -linearity and the coefficient multiplication, this determines the full ring.*

Proof. Use the two reductions

$$x_1^3 = a^2 x_1, \quad x_2^2 = -x_1 x_2 - x_1^2 + a^2 \quad (7.1)$$

from (6.3). For example, $A^2 = D - aA$, and

$$AB = x_1^2 + x_1 x_2 + ax_1 = C + D.$$

Substituting $x_2 = B - A - a$ in the second reduction gives $B^2 = C - aB$. Multiplication of $C = x_1 x_2$ by A gives $AC = F - aC$, while $AD = x_1^3 + ax_1^2 = aD$. For the other divisor products, reduction gives

$$BC = x_1^2 x_2 + x_1 x_2^2 + ax_1 x_2 = aC, \quad BD = F - aD.$$

Also $AF = BF = aF$, by the same reductions. The remaining entries follow from these formulas without any division:

$$\begin{aligned} C^2 &= C(B^2 + aB) = a^2 C + a^2 C = -a^2 C, \\ D^2 &= D(A^2 + aA) = -a^2 D, \\ CD &= (CA)(A + a) = (F - aC)(A + a) = aF, \\ CF &= C^2(A + a) = -a^2 F, \\ DF &= (DC)(A + a) = aF(A + a) = -a^2 F, \\ F^2 &= (FC)(A + a) = a^3 F. \end{aligned}$$

All signs are in $\mathbb{F}_3$. Each reduction is an equality in the genuine ring presentation, including when coefficients lie in N . $\square$

The three identities (1.4) are the divisor, or Monk–Chevalley-type, formulas in this example. All structure constants have the form ca^r with $c \in \mathbb{F}_3$ and $r \geq 0$. Indeed the coefficient of σ_z in $\sigma_v \sigma_w$ has degree $(\ell(v) + \ell(w) - \ell(z))\lambda$ and is constrained by Lemma 2.1. The negative cone contributes additional classes but no additional Schubert structure constants.

Table 2: Normalized Schubert lifts.

w	Notation	Degree	Representative
123	1	0	1
213	A	λ	x_1
132	B	λ	$x_1 + x_2 + a$
231	C	2λ	$x_1 x_2$
312	D	2λ	$x_1(x_1 + a)$
321	F	3λ	$x_1 x_2(x_1 + a)$

Table 3: Schubert multiplication; the lower triangle follows by commutativity.

$\cdot$	A	B	C	D	F
A	$D - aA$	$C + D$	$F - aC$	aD	aF
B		$C - aB$	aC	$F - aD$	aF
C			$-a^2C$	aF	$-a^2F$
D				$-a^2D$	$-a^2F$
F					a^3F

8 Fixed-point restriction and edge congruences

8.1 Restriction matrix and localization

At p_w , the line quotient L_i has character χ^{w_i-1} . Naturality of (6.1), and the coefficient comparison in degree λ , imply

$$x_i|_{p_w} = r_{w_i}a, \quad (r_1, r_2, r_3) = (0, -1, 1). \quad (8.1)$$

Substitution in Table 2 gives the restriction matrix in Table 4.

The diagonal entry at w is the normalized Euler product $q_R^{-1}(u^{\ell(w)}e_B(W_w))$, with the inverse taken in degree $\ell(w)\lambda$. These entries are $1, -a, -a, -a^2, -a^2, a^3$. For instance $W_{231} = \chi^2 \oplus \chi$ contributes $(-a)a = -a^2$, and $W_{321} = \chi^2 \oplus \chi \oplus \chi^2$ contributes a^3 .

With basis elements indexing its columns, the restriction matrix T factors as

$$T = M \operatorname{diag}(1, a, a, a^2, a^2, a^3), \quad M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 & 0 & 0 \\ 1 & 1 & -1 & 0 & -1 & 0 \\ 1 & 1 & 1 & -1 & -1 & 1 \end{pmatrix}. \quad (8.2)$$

The matrix M has determinant 1 in $\mathbb{F}_3$, whereas $\det T = a^9$.

Corollary 8.1. *Fixed-point restriction induces an isomorphism of graded rings*

$$H_G^*(X)[a^{-1}] \xrightarrow{\cong} \prod_{w \in S_3} R[a^{-1}].$$

Table 4: Restriction to the six fixed flags.

p_w	1	A	B	C	D	F
123	1	0	0	0	0	0
213	1	$-a$	0	0	0	0
132	1	0	$-a$	0	0	0
231	1	$-a$	a	$-a^2$	0	0
312	1	a	$-a$	0	$-a^2$	0
321	1	a	a	$-a^2$	$-a^2$	a^3

Proof. Both factors of (8.2) become invertible after inverting a . Restriction is a ring map, so the resulting module isomorphism is also a ring isomorphism. $\square$

8.2 The kernel and image before localization

Proposition 8.2. *At G/G , the kernel is*

$$\ker i^* = J_1A \oplus J_1B \oplus J_2C \oplus J_2D \oplus J_3F.$$

It is a nonzero square-zero ideal. In particular,

$$0 \neq \mu_{1,1}A \in H_G^2(X), \quad i^*(\mu_{1,1}A) = 0. \quad (8.3)$$

Proof. Invert M in (8.2), without inverting a . The kernel is that of the diagonal factor, giving the stated direct sum. Its coefficients lie in N by (2.6), so $N^2 = 0$ makes the kernel square zero. The class $\mu_{1,1}$ and the relation $a\mu_{1,1} = 0$ are part of (2.2)–(2.4), obtained from [2, Proposition 3.4]. Freeness of the Schubert basis makes $\mu_{1,1}A$ nonzero. Its degree is $(2 - \lambda) + \lambda = 2$. $\square$

Proposition 8.3. *A homogeneous tuple $f = (f_e, f_{s_1}, f_{s_2}, f_{231}, f_{312}, f_{321}) \in (R^\alpha)^6$ belongs to $\text{im } i^*$ if and only if*

$$\begin{aligned} f_e - f_{s_1} &\in aR, \\ f_e - f_{s_2} &\in aR, \\ f_e + f_{s_1} - f_{s_2} - f_{231} &\in a^2R, \\ f_e - f_{s_1} + f_{s_2} - f_{312} &\in a^2R, \\ f_{321} - f_e + f_{s_1} + f_{s_2} - f_{231} - f_{312} &\in a^3R. \end{aligned} \quad (8.4)$$

Membership in $a^r R$ here means membership in $a^r R^{\alpha-r\lambda}$.

Proof. Direct inversion of M gives

$$M^{-1}f = \begin{pmatrix} f_e \\ f_e - f_{s_1} \\ f_e - f_{s_2} \\ f_e + f_{s_1} - f_{s_2} - f_{231} \\ f_e - f_{s_1} + f_{s_2} - f_{312} \\ f_{321} - f_e + f_{s_1} + f_{s_2} - f_{231} - f_{312} \end{pmatrix}.$$

The tuple lies in the image of T exactly when these coordinates are divisible by $1, a, a, a^2, a^2, a^3$, respectively. Conversely, choose preimages under the corresponding powers of a as Schubert coefficients. Different choices differ by an element of the kernel in Proposition 8.2. $\square$

8.3 Root curves

For each transposition (ij) and flag p_w , the subgroup acting unitarily on $E_{w_i} \oplus E_{w_j}$ produces a root curve $\mathbb{CP}^1$ joining p_w to $p_{w(ij)}$. Its tangent character at p_w is $\chi^{w_j - w_i}$, and the character at the other end is its inverse. There are nine such curves. These are the torus-invariant root curves, not an enumeration of all curves invariant under the finite group.

Each root curve is equivariantly a representation sphere S^λ , up to the orientation choice. Its two fixed-point restrictions have the form $(c, c \pm ad)$. This follows from $H_G^\alpha(S^\lambda) \cong R^\alpha \oplus R^{\alpha-\lambda}$: the

reduced generator restricts to zero at the basepoint and to the Euler class at the other fixed point. Thus the condition from every root curve is

$$f_v - f_w \in aR. \quad (8.5)$$

The ideals generated by a and $-a$ agree.

Corollary 8.4. *Let $\Gamma \subset R^6$ be the graded R -module of tuples satisfying (8.5) on all nine root curves. There is a graded R -module isomorphism*

$$\Gamma / \text{im } i^* \cong \left(\frac{aR}{a^2R} \right)^{\oplus 2} \oplus \frac{aR}{a^3R}. \quad (8.6)$$

The quotients carry their inherited gradings as quotients of ideals in R , with no additional shift. In particular the edge conditions do not determine $\text{im } i^$.*

Proof. The root graph is the Cayley graph of S_3 on its three transpositions, hence is connected. Since every edge ideal is aR , a tuple belongs to Γ exactly when all its entries agree modulo aR . The first column of M is the constant vector, so

$$M^{-1}\Gamma = R \oplus (aR)^{\oplus 5}, \quad M^{-1}(\text{im } i^*) = R \oplus (aR)^{\oplus 2} \oplus (a^2R)^{\oplus 2} \oplus a^3R.$$

Taking the coordinatewise quotient proves (8.6). For $\xi = M^{-1}f$, with coordinates numbered 0 through 5, the quotient map is

$$f \mapsto (\xi_3 \bmod a^2R, \xi_4 \bmod a^2R, \xi_5 \bmod a^3R).$$

It is surjective: for any representatives $b, c, d \in aR$, the tuple $M(0, 0, 0, b, c, d)^t$ belongs to Γ and maps to their classes. Its kernel is precisely the image described in Proposition 8.3. All maps preserve representation degree, and no power of a has been cancelled.

For example, take

$$f = (0, 0, 0, 0, 0, a) \in (R^\lambda)^6.$$

It lies in Γ , and $M^{-1}f = f$. Its last coordinate gives the nonzero class of a in aR/a^3R : a homogeneous preimage under multiplication by a^3 would lie in $R^{-2\lambda} = 0$. $\square$

In pure multiples of λ , restriction itself is injective: Schubert coefficients lie in $\mathbb{F}_3[a]$, on which the diagonal factor of (8.2) is injective. The higher divisibilities in Corollary 8.4 constrain the image even in these pure degrees. Conversely, $a^r N = N$ for every $r \geq 1$, so Proposition 8.3 gives $N^6 \subset \text{im } i^*$. Thus negative-cone tuples themselves impose no higher-divisibility obstruction.

Harada–Henriques–Holm’s injectivity theorem requires non-zero-divisor Euler classes [6, Theorem 2.3]; here $a\mu_{1,1} = 0$ with $\mu_{1,1} \neq 0$. Their image theorem additionally requires the relative-primality condition that divisibility by each Euler factor imply divisibility by their product [6, Assumption 4 and Theorem 3.1]. For a two-dimensional complex cell the factors here are $\pm a$, and $(a) \cap (a) = (a) \neq (a^2)$: indeed $a \notin a^2R$ since $R^{-\lambda} = 0$. This second failure already occurs in the pure representation degrees, where the coefficient classes lie in $\mathbb{F}_3[a]$. The classical torus descriptions over $\mathbb{R}$ [5, Theorem 1.2.2] and $\mathbb{Q}$ [8, §2.3] concern Borel cohomology. They do not directly give an edge presentation in this genuine C_3 -equivariant setting.

9 Mackey structure and consistency checks

9.1 Coefficient restriction and transfer

At the trivial orbit the coefficient ring is

$$\underline{R}^*(G/e) = \mathbb{F}_3[v^{\pm 1}], \quad |v| = \lambda - 2.$$

In a particular degree α this notation records the ordinary coefficient group in dimension $|\alpha|$; distinct powers of v are distinct $RO(G)$ degrees of underlying dimension zero. Restriction is

$$\text{res}(u) = v, \quad \text{res}(a) = \text{res}(\kappa) = 0, \quad \text{res}(N) = 0. \quad (9.1)$$

For a and κ this follows from the underlying coefficient degrees 2 and 1. For $\nu_{j,k}$ the underlying degree is $1 - 2k$, and for $\mu_{j,k}$ it is $2 - 2k$; only $\mu_{j,1}$ could have nonzero restriction. The following sphere calculation shows that its restriction is zero.

Lemma 9.1. *The negative-cone generators can be normalized compatibly with (2.4) so that*

$$\text{tr}(v^n) = \begin{cases} 0, & n \geq 0, \\ \mu_{-n,1}, & n < 0. \end{cases} \quad (9.2)$$

Proof. For $n \geq 0$, Frobenius reciprocity and $\text{tr}(1) = 3 = 0$ give $\text{tr}(v^n) = u^n \text{tr}(1) = 0$. For $j \geq 1$, use the cofiber sequence

$$S(j\lambda)_+ \longrightarrow S^0 \longrightarrow S^{j\lambda}.$$

Its cohomology connecting map is an isomorphism in degree $2j - 1$ to reduced degree $2j$ on $S^{j\lambda}$, both at G/G and at G/e , since the adjacent positive integer coefficients of a point vanish. The sphere $S(j\lambda)$ is free, with quotient the oriented lens space $L^{2j-1}(3)$. Integer-graded cohomology with constant coefficients agrees here with ordinary cohomology of that quotient [2, Proposition 3.2]. On this free space, restriction and transfer are the pullback and cohomological transfer of the three-sheeted covering

$$S^{2j-1} \longrightarrow L^{2j-1}(3).$$

The covering transfer is induced by summing the lifts of singular simplices [7, §3.G, pp. 321–322]. On top integral cohomology pullback is multiplication by 3, whereas transfer sends the sphere's orientation class to that of the lens space: evaluation of a transferred cocycle on the base fundamental class equals evaluation on its lifted fundamental class. Reducing modulo 3, pullback is zero and transfer is nonzero. Naturality of the connecting maps gives the same assertion on $\widetilde{H}_G^{2j}(S^{j\lambda})$.

Desuspending by $j\lambda$ gives target degree $2j - j\lambda$, with $R^{2j-j\lambda} = \mathbb{F}_3\mu_{j,1}$, and underlying generator v^{-j} . Thus (9.1) holds and

$$\text{tr}(v^{-j}) = c_j \mu_{j,1}, \quad c_j \in \mathbb{F}_3^\times.$$

For $j > 1$, Frobenius reciprocity gives $u \text{tr}(v^{-j}) = \text{tr}(v^{1-j})$. Since $u\mu_{j,1} = \mu_{j-1,1}$, we obtain $c_j = c_{j-1}$. Writing c for this common scalar, replace every $\nu_{j,k}$ and $\mu_{j,k}$ by $c\nu_{j,k}$ and $c\mu_{j,k}$, respectively. This preserves all the relations in (2.4) and gives (9.2). $\square$

The vanishing of $\text{tr}(1)$ does not imply that transfers vanish in all representation degrees.

9.2 The flag cohomology as a graded Green functor

Let $y_i = c_1(L_i^\vee) \in H^2(X; \mathbb{F}_3)$. The value at G/e is

$$\underline{H}_G^*(X)(G/e) = \frac{\mathbb{F}_3[v^{\pm 1}, y_1, y_2, y_3]}{(e_1(y), e_2(y), e_3(y))}, \quad |y_i| = 2, \quad |v| = \lambda - 2. \quad (9.3)$$

Restriction on coefficients is (9.1), and on the ring generators it is $\text{res}(x_i) = vy_i$. If $\mathfrak{S}_w(y)$ denotes the ordinary Schubert polynomial, define

$$\bar{\sigma}_w = \text{res}(\sigma_w) = v^{\ell(w)} \mathfrak{S}_w(y).$$

This bar notation includes the displayed power of v . Then

$$\text{tr}(v^n \bar{\sigma}_w) = \begin{cases} 0, & n \geq 0, \\ \mu_{-n,1} \sigma_w, & n < 0. \end{cases} \quad (9.4)$$

Indeed Frobenius reciprocity gives $\text{tr}(b \text{res} \sigma_w) = \text{tr}(b) \sigma_w$, and Lemma 9.1 applies. Since the $\bar{\sigma}_w$ form a basis over $\mathbb{F}_3[v^{\pm 1}]$, this specifies every transfer.

The Mackey Weyl action of $W_G(e) = G$ on (9.3) is trivial: the action on X comes from a connected torus and is homotopic to the identity, and G preserves all representation orientations. The Mackey double-coset formula consequently reads

$$\text{res tr} = 1 + g + g^2 = 0,$$

as also follows from (9.1) and (9.4). For Frobenius reciprocity, terms in the Schubert multiplication involving a positive power of a vanish both upon restriction and upon multiplication by $\mu_{j,1}$. The remaining terms are the ordinary Schubert products. Coefficient reciprocity also holds on the generators: $u\mu_{j,1} = \mu_{j-1,1}$, $a\mu_{j,1} = \kappa\mu_{j,1} = 0$, and $N\mu_{j,1} = 0$. These identities verify Frobenius reciprocity on an additive basis, hence on all elements. Together with Theorem 1.1, they specify the graded Green functor.

The G/e value of fixed-point restriction sends every y_i to zero and sends v to the same v in all six factors. Its image is the diagonal copy of $\mathbb{F}_3[v^{\pm 1}]$ and its kernel is (y_1, y_2, y_3) . This lower-orbit kernel is not square zero; for example $y_1^2 \neq 0$. Thus the square-zero assertion of Proposition 8.2 applies only at G/G .

9.3 Ordinary specialization and the top class

Forgetting the action and the representation grading sends a, κ, N to zero and sends u, v to the ordinary unit. It gives

$$H^*(X; \mathbb{F}_3) \cong \mathbb{F}_3[y_1, y_2, y_3]/(e_1(y), e_2(y), e_3(y)).$$

The basis has ordinary degrees 0, 2, 2, 4, 4, 6 and rank $6 = |S_3|$. Restriction sends F to $v^3 y_1^2 y_2$. On the bundle $\text{Fl}(\mathbb{C}^3) \rightarrow \mathbb{CP}^2$ selecting the first line, y_2 integrates to 1 on the $\mathbb{CP}^1$ fiber and y_1 is the positive hyperplane class on the base. Therefore

$$\int_X y_1^2 y_2 = 1.$$

So F specializes to the top Schubert class with the complex orientation. Finally, Table 3 gives

$$A^2 B = F + a(D - C), \quad AB^2 = F + a(C - D).$$

After setting $v = 1$, the underlying classes satisfy $\bar{A}^2 \bar{B} = \bar{A} \bar{B}^2 = \bar{F}$. The genuine formula $F^2 = a^3 F$ is also consistent with its restriction a^3 at 321 and zero at the other five flags.

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